Name:- Divyanka Sagolale
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```
div12@div12:~/CC lab$ yacc -d desk.y
div12@div12:~/CC lab$ lex desk.l
div12@div12:~/CC lab$ gcc y.tab.c lex.yy.c -o desk
div12@div12:~/CC lab$ ./desk
Desk Calculator
63/5
Result = 12
3+*2
Invalid expression
```

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```
%{
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
int yylex(void);
int yyerror(const char *s);
%}

%token NUM INC DEC
%left '+' '-'
%left '*' '/' '%'
%right '^'
%left INC DEC
%%
input:
    | input line
    ;

line:
      '\n'
    | expr '\n'  { printf("Result = %d\n", $1); }
    | error '\n' { printf("Invalid expression\n"); yyerrok; }
    ;

expr:
      expr '+' expr     { $$ = $1 + $3; }
    | expr '-' expr     { $$ = $1 - $3; }
    | expr '*' expr     { $$ = $1 * $3; }
    | expr '/' expr     { $$ = $1 / $3; }
    | expr '%' expr     { $$ = $1 % $3; }
    | expr '^' expr     { $$ = 1;
                          for (int i = 0; i < $3; i++)
                              $$ *= $1;
                        }
    | expr INC          { $$ = $1 + 1; }
    | expr DEC          { $$ = $1 - 1; }
    | '(' expr ')'      { $$ = $2; }
    | NUM
    ;

%%
int main(void)
{
    printf("Desk Calculator\n");
    return yyparse();
}

int yyerror(const char *s)
{
    return 0;
}
```

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```
%{
#include <stdio.h>
#include <stdlib.h>
#include "y.tab.h"
%}

%%

[0-9]+       { yylval = atoi(yytext); return NUM; }
"++"         { return INC; }
"--"         { return DEC; }
[ \t]+       ;
\n           { return '\n'; }
.            { return yytext[0]; }

%%

int yywrap(void)
{
    return 1;
}
```